

Lab2 : Nginx Web Server

Question 1: Install Nginx

```
sudo apt update
sudo apt install nginx -y
sudo systemctl status nginx
```

```
nagy@ZenBook-UX425EA:~$
nagy@ZenBook-UX425EA:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-03-09 22:12:57 EET; 24s ago
     Docs: man:nginx(8)
  Process: 47747 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exi>
  Process: 47748 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, st>
 Main PID: 47749 (nginx)
    Tasks: 9 (limit: 18718)
   Memory: 8.7M
      CPU: 31ms
   CGroup: /system.slice/nginx.service
           └─47749 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─47750 "nginx: worker process"
               └─47751 "nginx: worker process"
                 └─47753 "nginx: worker process"
                   └─47754 "nginx: worker process"
                     └─47755 "nginx: worker process"
                       └─47756 "nginx: worker process"
                         └─47757 "nginx: worker process"
                           └─47758 "nginx: worker process"

Mar 09 22:12:57 ZenBook-UX425EA systemd[1]: Starting A high performance web server and a rever>
Mar 09 22:12:57 ZenBook-UX425EA systemd[1]: Started A high performance web server and a revers>
nagy@ZenBook-UX425EA:~$
nagy@ZenBook-UX425EA:~$
```

Question 2: Change the default file (index.html) to (yourName.html)

```
cd /var/www/html
sudo mv index.html john.html
sudo nano /etc/nginx/sites-available/default
sudo nginx -t
sudo systemctl reload nginx
```

```
nagy@ZenBook-UX425EA:~$
nagy@ZenBook-UX425EA:~$
nagy@ZenBook-UX425EA:~$ cd /var/www/html
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo mv index.html nagy.html
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo nano /etc/nginx/sites-available/default
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo systemctl reload nginx
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ curl localhost
<h2>This is new html</h2>
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$
```

Question 3: Make two html files, and change the configuration file to access the first file on port 81, and access the second file on port 82.

```
sudo mkdir -p /var/www/lab/site1 /var/www/lab/site2
echo "<h1>Welcome to Port 81</h1>" | sudo tee /var/www/lab/site1/index.html
echo "<h1>Welcome to Port 82</h1>" | sudo tee /var/www/lab/site2/index.html
sudo nano /etc/nginx/sites-available/multi_port
sudo ln -s /etc/nginx/sites-available/multi_port /etc/nginx/sites-enabled/
sudo nginx -t
sudo systemctl restart nginx
```

```
nagy@ZenBook-UX425EA: /var/www/html$
nagy@ZenBook-UX425EA: /var/www/html$ sudo mkdir -p /var/www/lab/site1 /var/www/lab/site2
nagy@ZenBook-UX425EA: /var/www/html$
nagy@ZenBook-UX425EA: /var/www/html$ echo "<h1>Welcome to Port 81</h1>" | sudo tee /var/www/lab/
site1/index.html
<h1>Welcome to Port 81</h1>
nagy@ZenBook-UX425EA: /var/www/html$
nagy@ZenBook-UX425EA: /var/www/html$ echo "<h1>Welcome to Port 82</h1>" | sudo tee /var/www/lab/
site2/index.html
<h1>Welcome to Port 82</h1>
nagy@ZenBook-UX425EA: /var/www/html$
nagy@ZenBook-UX425EA: /var/www/html$
nagy@ZenBook-UX425EA: /var/www/html$ sudo nano /etc/nginx/sites-available/multi_port
nagy@ZenBook-UX425EA: /var/www/html$
nagy@ZenBook-UX425EA: /var/www/html$ sudo ln -s /etc/nginx/sites-available/multi_port /etc/nginx
/sites-enabled/
nagy@ZenBook-UX425EA: /var/www/html$
nagy@ZenBook-UX425EA: /var/www/html$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
nagy@ZenBook-UX425EA: /var/www/html$
nagy@ZenBook-UX425EA: /var/www/html$ sudo systemctl restart nginx
nagy@ZenBook-UX425EA: /var/www/html$
nagy@ZenBook-UX425EA: /var/www/html$
nagy@ZenBook-UX425EA: /var/www/html$ curl localhost:81
<h1>Welcome to Port 81</h1>
nagy@ZenBook-UX425EA: /var/www/html$
nagy@ZenBook-UX425EA: /var/www/html$ curl localhost:82
<h1>Welcome to Port 82</h1>
nagy@ZenBook-UX425EA: /var/www/html$
nagy@ZenBook-UX425EA: /var/www/html$
```

Question 4: Main differences between Apache & Nginx

Feature	Apache	Nginx
Architecture	Process-based (creates a new thread for every request).	Event-driven (Asynchronous, handles multiple connections in one thread).
Performance	Slower under heavy loads due to high RAM consumption.	High performance and stability under heavy concurrent loads.
Static Content	Good, but slightly slower than Nginx.	Excellent at serving static files (much faster).
Dynamic Content	Handles dynamic content within the server (via modules).	Requires an external processor (like PHP-FPM) to handle dynamic content.
Configuration	Allows directory-level configuration via <code>.htaccess</code> files.	Centralized configuration only (better for performance).